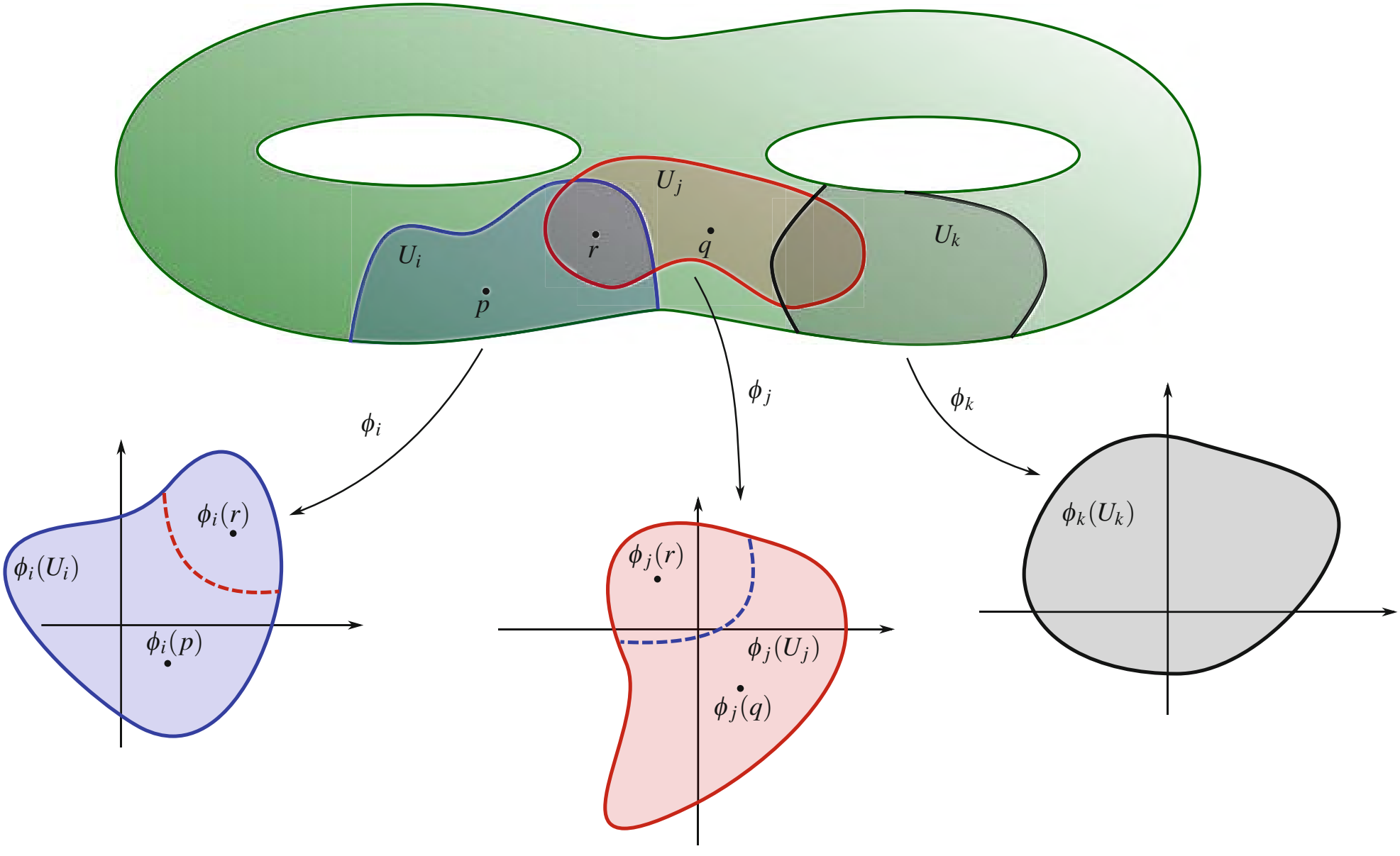

U_i
U_j
U_k
p
r
q
ϕ_i
ϕ_j
ϕ_k
$\phi_i(U_i)$
$\phi_i(r)$
$\phi_i(p)$
$\phi_j(r)$
$\phi_j(U_j)$
$\phi_j(q)$
$\phi_k(U_k)$